

SUMMARY

AI/ML Engineer with around 4 years of experience building and deploying scalable machine learning and Generative AI solutions across cloud-native environments. Experienced in developing NLP, computer vision, and RAG-based applications using Python, PyTorch, Hugging Face, LangChain, and FastAPI. Strong expertise in model deployment, MLOps, vector search, and inference optimization using Kubernetes, Docker, TensorRT, and MLflow. Skilled in building end-to-end ML pipelines, integrating AI services into enterprise applications, and improving deployment reliability through monitoring, CI/CD, and observability frameworks across AWS and Azure platforms.

TECHNICAL SKILLS

Programming Languages	Python, SQL, Bash, Java
Machine Learning & Deep Learning	PyTorch, TensorFlow, Scikit-learn, XGBoost, LightGBM, Hugging Face Transformers, PEFT, LLaMA 2/3, GPT, T5, FLAN-T5, RoBERTa, BERT, CLIP
Generative AI & LLMs	LangChain, LlamaIndex, Whisper, RAG Pipelines, Prompt Engineering, QLoRA, LoRA, Adapters
NLP & Conversational AI	spaCy, NLTK, Named Entity Recognition (NER), Sentiment Analysis, Text Classification, Document Parsing, OCR (Tesseract, AWS Textract)
Model Serving & Optimization	TorchServe, ONNX, ONNX Runtime, TensorRT, TorchScript, Triton Inference Server, Mixed-Precision Quantization (FP16/INT8), FastAPI, Flask, Streamlit
Vector Search & Semantic Retrieval	FAISS, Pinecone, Semantic Search
MLOps & CI/CD	MLflow, Airflow, DVC, Docker, Kubernetes, Terraform, GitHub Actions, Jenkins, AWS CodePipeline, Lambda, Experiment Tracking
Real-Time Data & Pipelines	Apache Spark, PySpark, Kafka (Confluent), ETL Pipelines, dbt, Hive, Feature Stores
Cloud Platforms & Infrastructure	AWS (SageMaker, Lambda, S3, ECS, ECR, CloudWatch), GCP (Vertex AI, AI Platform), Azure ML Studio, AWS Bedrock
Responsible AI & Safety	SHAP, LIME, Guardrails AI, Presidio, Prompt Injection Protection, Human-in-the-Loop Systems, GDPR/RBI/AML/KYC Compliance
Version Control & Collaboration	Git, GitHub, GitLab, Pre-commit Hooks, Agile/Scrum, Code Reviews, A/B Testing
Data Visualization & Analytics	Tableau, Power BI, Matplotlib, Business Intelligence Dashboards, Error Analysis Visuals

PROFESSIONAL EXPERIENCE

Cigna HealthCare
AI/ML Engineer

Jan 2026 – Present

- Built scalable inference APIs using FastAPI and optimized PyTorch models using TensorRT for low-latency predictions.
- Fine-tuned LLMs using LoRA/adapters and implemented RAG pipelines using LangChain and FAISS, improving retrieval quality and response performance.
- Automated ML deployment workflows using Kubernetes and CI/CD pipelines, improving deployment consistency across environments.
- Owned end-to-end deployment of NLP and LLM models from development to production across Kubernetes-based environments.
- Investigated and fixed pipeline failures impacting downstream analytics workflows, reducing incident resolution time.
- Implemented model versioning, regression validation, and containerized rollouts, ensuring safe deployment of

multiple LLMs, speech, and vision models across GPU clusters.

- Implemented monitoring and observability solutions using Prometheus and CloudWatch to improve deployment reliability and SLA tracking.
- Collaborated with cross-functional teams to integrate microservices with AWS IAM-based access control, ensuring secure multi- cloud deployments.
- Integrated ML inference APIs with internal enterprise applications to support real-time predictions and search capabilities.
- Mentored engineers on GPU optimization, distributed training, and containerized AI microservices, fostering team-wide technical excellence.

United Airlines
AI/ML Engineer

July 2024 - Dec 2025

- Performed comprehensive data quality checks, profiling, and validation on clinical and claims data to ensure accuracy, completeness, and consistency across products.
- Implemented automated data quality frameworks to proactively detect anomalies and maintain high trust in downstream analytics.
- Managed scalable ETL pipelines processing structured and semi-structured data from APIs and cloud storage systems.
- Built end-to-end ML pipelines with automated training, validation, and deployment workflows in AWS.
- Monitored model performance using precision/recall, ROC-AUC, and implemented drift detection mechanisms.
- Implemented ETL workflows with Airflow/Glue to cleanse, transform, and standardize data for downstream ML models.
- Engineered ML features from operational logs (resolution states, escalation types, latency metrics).
- Applied validation rules and anomaly detection models to improve dataset quality and proactively flag issues.
- Built scalable data ingestion pipelines from REST APIs, flat files (CSV/JSON), and cloud storage (S3) using Python and Spark.
- Built observability dashboards to monitor inference latency, failures, and deployment health across production systems.
- Designed and optimized Snowflake warehouses, implemented RBAC policies, and improved cost efficiency through query and compute optimization.
- Implemented metadata tracking, lineage documentation, and observability frameworks using dbt tests and Great Expectations to ensure pipeline reliability.
- Prototyped classification models (XGBoost, Random Forest) to predict resolution outcomes and escalation risks.
- Evaluated models using precision/recall, ROC-AUC, and monitored drift across time slices.
- Published curated datasets into Redshift and Parquet tables for analytics dashboards (Tableau/Power BI).
- Authored documentation and SOPs for reproducible workflows; trained stakeholders on self-service reporting.
- Reduced manual incident triage by 20% and improved anomaly detection coverage during pilot deployment.

NICE Systems Ltd
AI/ML Engineer

Sep 2022 - Nov 2023

- Designed and developed computer vision pipelines for driver monitoring applications using PyTorch and Transformer-based models.
- Built scalable video processing pipelines using Python, Spark, and Airflow for automated preprocessing and feature extraction.
- Optimized and deployed vision models using ONNX and TensorRT for low-latency edge inference.
- Implemented experiment tracking and model lifecycle management using MLflow and Optuna.
- Developed semantic video search systems using FAISS and vector embeddings to improve retrieval efficiency.
- Applied explainable AI techniques including SHAP and LIME to improve model interpretability and fairness.
- Designed and executed large-scale A/B testing and performance evaluations, improving fairness, reducing false positives by 20%, and increasing user trust and adoption among state transport authorities.
- Collaborated with cross-functional teams including government stakeholders, product managers, and research scientists to scale deployments and contribute to open-source computer vision datasets and models.
- Implemented MLOps best practices for continuous deployment, monitoring, and versioning using Git, Python-based

automation, CI/CD pipelines, and Azure DevOps, reducing deployment failures by 30%.

EDUCATION

- Masters in Data Science and Artificial Intelligence
- Bachelors in Computer Science with Specialization in Data Analytics